

Would you risk everything on a coin flip if the math told you to?

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Norway, 2026. Captured by Trent Conley

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Imagine you have \$1 and a stranger has a fair coin. They offer you the chance to play a game 100 times, and the rules are as follows:

1. You can bet any amount each round.
2. If it lands heads, you get your bet back plus 2 times your bet amount.
3. If it lands tails, you lose your bet.

For example, betting \$1 on the first round results in you having \$3 (heads) or \$0 (tails).

Do you play? Assuming we trust this mysterious stranger, I think most people would say yes. But how much should you bet? If you bet everything for each 100 rounds, you'd almost certainly walk away a dollar poorer.

This question intrigued me, because I saw two competing concepts: expected value and median outcome. To summarize shortly, expected outcome is the weighted probability sum of an outcome and its respective payout, and median is the middle outcome.

A quick expected value calculation will tell you that you should bet all of your money every time. But is that the right amount to bet? What is the probability of profiting?

If you bet everything, you need to get heads at every turn to realize a profit. The probability of that is 1 in 2^{100} , which is essentially zero (roughly one in a billion times a billion times a billion times a thousand, so yeah pretty small).

So clearly, we shouldn't bet everything if we want to actually realize a profit. Put another way, I only care about the median outcome. Before I go into the math to prove it, take a guess. 80%? 50%? 20%?

First, let's define everything more formally:

h = number of heads

b = bet fraction

M_i = initial money

M_f = final money

Then, we can define your final bankroll as:

$$M_f = M_i(1 + 2b)^h(1 - b)^{100-h}$$

Great. Now what does the median outcome look like? I'd argue that it is when there are 50 heads and 50 tails out of intuition, because this sits at exactly the middle distribution of outcomes. This will get more tricky later, but for this case is rather simple.

So, our median outcome final money is:

median $h = 50$

$$M_f = M_i(1 + 2b)^{50}(1 - b)^{50}$$

We now have something to optimize. Let's take the derivative of M_f with respect to b .

$$\begin{aligned}
\frac{dM_f}{db} &= M_i \left[100(1+2b)^{49}(1-b)^{50} \right. \\
&\quad \left. - 50(1+2b)^{50}(1-b)^{49} \right] \\
&= 50M_i(1+2b)^{49}(1-b)^{49}(1-4b)
\end{aligned}$$

Now, let's set it to zero to find our critical points.

$$0 = 50M_i(1+2b)^{49}(1-b)^{49}(1-4b)$$

$$b \in \left\{ -\frac{1}{2}, \frac{1}{4}, 1 \right\}$$

b can't be negative, and if it is 1, then our final money is zero, so it must be 0.25! We need to bet 25% of our stack for each of the 100 rounds.

I think we have a sufficient answer, but the interesting part to me is a step further: what if you could run multiple of these games in parallel?

Let's first simplify the problem, and only consider 10 rounds. The probability of betting on just heads becomes 1 in 1024 which seems small, but it is much more attainable than our prior bet-everything-and-hope strategy. If we had, say 10,000 independent rounds played in parallel, you would expect that we should have at least one of the all heads rounds hit. More concretely, the probability of winning anything with the simultaneous rounds is $1 - .999^{10000} \approx 99.99\%$. Pretty good!

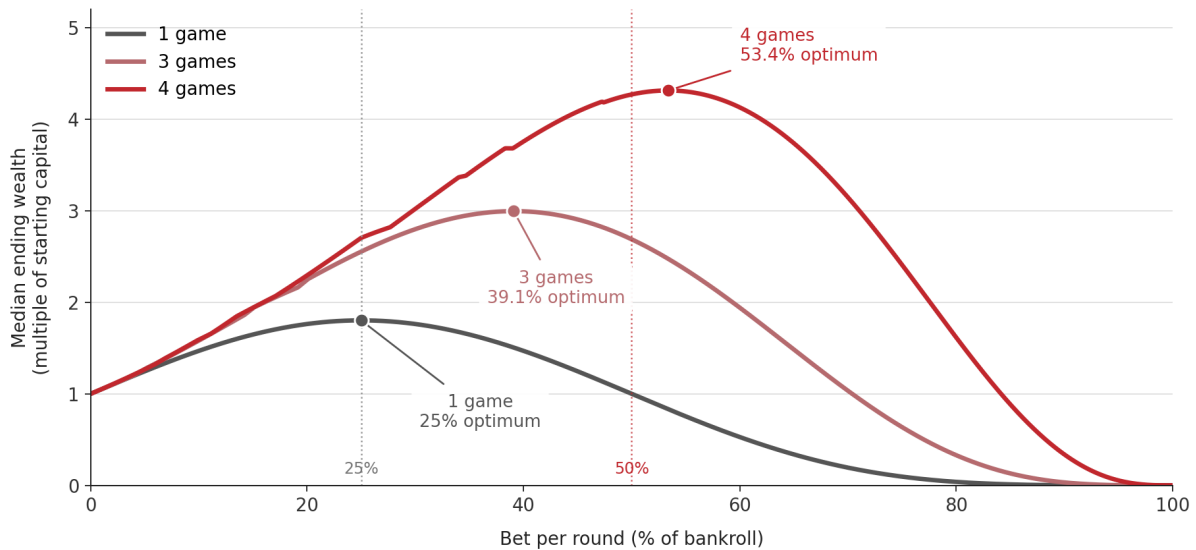
When we have enough independent rounds, we should bet everything because now the median outcome will contain multiple large winnings.

Okay, so we now know that if we have many simulations compared to rounds, we should bet everything. My natural questions that follow: when does betting 50% make sense? Is there some ratio between rounds and number of games, or does it change with the rounds? How do you model the bets in between 25% and 100%?

These questions intrigue me much more than the original thought experiment.

I ran a Monte Carlo simulation to find how many simultaneous games necessitates a 50% betting optimum, given our 10 rounds total.

Four games are enough to justify a 50% bet



Monte Carlo: 500,000 portfolios per curve, seed 20260726. The original capital is divided equally among the games.

At just four games, you should bet over 50%. That's a surprisingly small amount of parallel games. Just a few extra games completely change your strategy.

I found this variant of the question more analogous to real life. Venture capital firms expect most of their investments to go to zero, but the returns from a few outliers that become billion dollar companies cover the entire fund. The same goes for stocks: the majority of returns from the S&P 500 came from just seven companies between 2023 and 2025.

This is how capitalism, evolution, and nature work, through stochastic betting. The only question you need to ask yourself: how many games are you playing?

All of my code is open sourced on Github, along with a more well formatted version of the post created with LaTeX and rendered as a PDF. I'm a bit old school when writing, so these are all my words. All the code was generated.

If you see something that's cool, make a PR and I could feature it in the next post (such as what happens if the games are no longer independent!).

Also, for those scrutinizing the application of this to Venture capital and other domains, I've left out quite an important detail: most events are not independent. I'd challenge the reader to think about how this affects your strategy.